14. Write a code using Raw sockets to implement packet sniffing.

3/10/24

Aim:

Write a code using Raw sockets to implement packet sniffing

Code:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dest_ip = ip_layer.dst

# determine the protocol:
    protocol_name = ""
    if protocol == 1:
        protocol_name = "ICMP"
    elif protocol_name == 6:
        protocol_name = "TCP"
    elif protocol == 17:
        protocol_name = "UDP"
```

```
else
    protocol_name = "unknown protocol"

# print packet details.
print(f"protocol : {protocol_name}")
print(f"Source IP {src_ip}")
print(f"Destination ip : {dst_ip}")
print("_" * 50)

sniff(iface = "Wi-fi", prn = packet_callback, filter=
    store).
```

output:

```
protocol : TCP
Source IP : 57.155.141.17
Destination IP : 192.168.1.6
protocol : UDP
Source ip : 192.168.1.6
```

Result:

The implementation of packets using Raw Sockets. implemented sucessfully.

